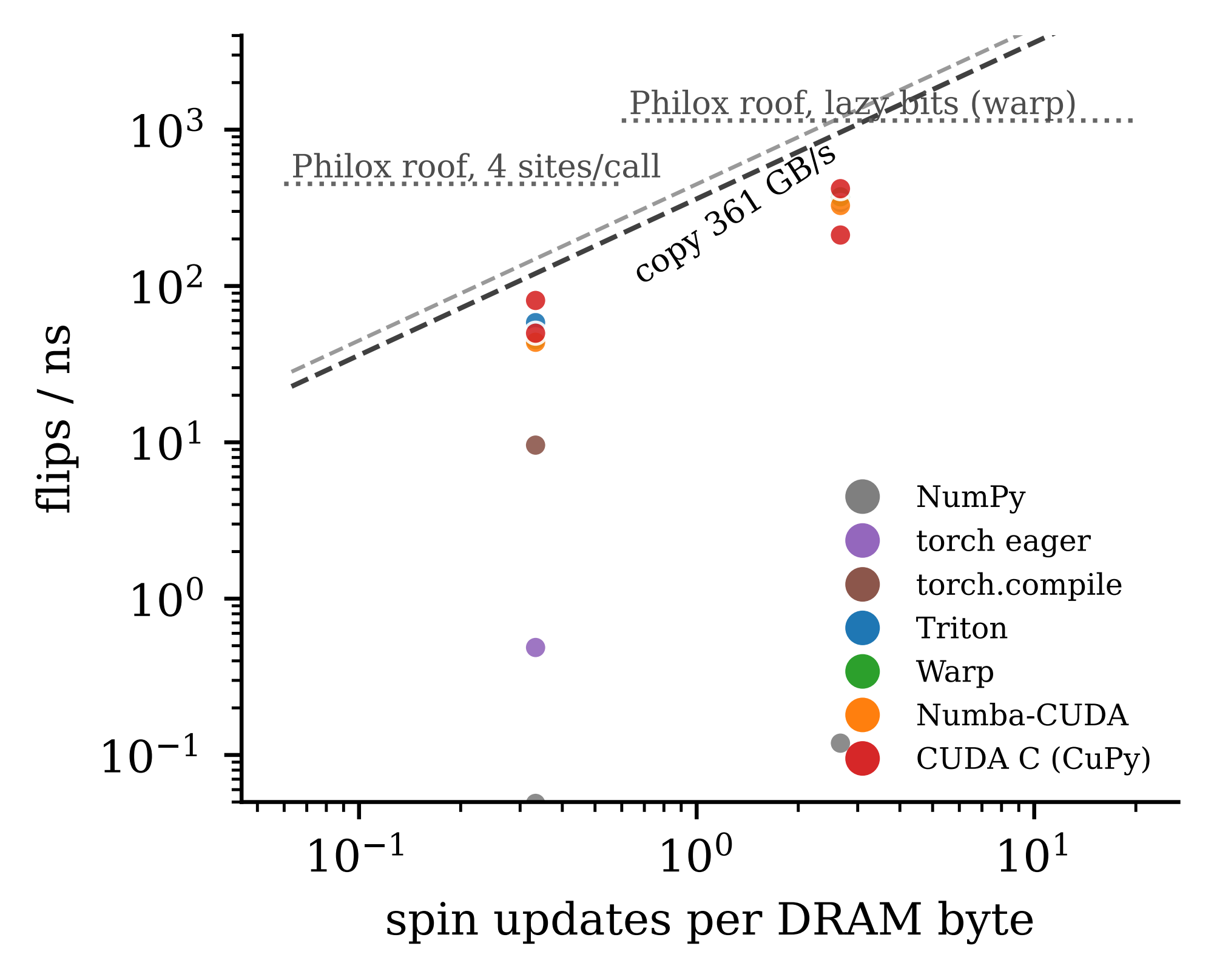
Philox roof, lazy bits (warp)
Philox roof, 4 sites/call
copy 361 GB/s
NumPy
torch eager
torch.compile
Triton
Warp
Numba-CUDA
CUDA C (CuPy)
flips / ns
spin updates per DRAM byte
$10^{3}$
$10^{2}$
$10^{1}$
$10^{0}$
$10^{-1}$
$10^{-1}$
$10^{0}$
$10^{1}$